

Direct Observation Method for Modular Inverses

- **Direct Observation Method:** This method involves testing each integer x from 1 to $m-1$, computing $ax(\bmod m)$, and identifying the value of x that yields a remainder of 1. Such an x satisfies the $ax(\bmod m)=1$ and is the modular inverse of $a \bmod m$.
- **Example:** Find the modular inverse of 5 modulo 7.
 - **Solution 1:** Using the *direct observation method* to find an integer x such that $5x \equiv 1(\bmod 7)$.
 - ① Try values of x from 1 to 6. When $x=3$, $5 \times 3(\bmod 7)=1$, satisfied $5x \equiv 1(\bmod 7)$.
 - ② Verify that $x=3$ is correct: $5^{-1} \equiv 3(\bmod 7)$.

↳ Trial Method for Modular Inverses (e.g.)

- Solution 2: Use the *trial method* to solve the congruence $ax \equiv 1 \pmod{m}$ and find the modular inverse of $a \pmod{m}$.
 - ① Since $\gcd(5, 7) = 1$, the modular inverse of 5 modulo 7 exists.
 - ② solve $ax \equiv 1 \pmod{m}$, try values of x (1 to 6) compute $5x \pmod{7}$.
 - ③ when $x = 3$, $5 \times 3 = 15 \pmod{7} = 1$.
 - ④ Therefore, the modular inverse of 5 modulo 7 is 3, that is, $5^{-1} \equiv 3 \pmod{7}$.

↳ Trial Method for Modular Inverses (e.g.)

- Solution 3: Use the *Euclidean Algorithm*. Since $\gcd(5,7)=1$, we solve the equation $5x + 7y = 1$.
 - ① $7 = 1 \times 5 + 2, 5 = 2 \times 2 + 1$, then $1 = 5 - 2 \times 2$.
 - ② $2 = 7 - 1 \times 5, 1 = 5 - 2 \times 2$, we have $3 \times 5 - 2 \times 7 = 1$, which is the modular inverse of 5 modulo 7, then $5 \times 3 \equiv 1 \pmod{7}$ holds.

■ 8.4.1 Linear Congruences

Modular Inverses

■ 8.4.2 The Chinese Remainder Theorem

■ 8.4.3 Arithmetic Operations with Large Integers

↳ Sunzi Suanjing and the Chinese Remainder Theorem

- 《孙子算经》 “物不知数” 问题: 今有物, 不知其数, 三三数之剩二, 五五数之剩三, 七七数之剩二, 问物几何?

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

- Theorem 8.14 (Chinese Remainder Theorem).

Let $m_1, m_2, \dots, m_k$ be pairwise coprime positive integers. Then the system of linear congruences $x \equiv a_i \pmod{m_i}$, $i=1, 2, \dots, k$ has an integer solution, and the solution is unique modulo, that is, any two solutions are congruent modulo m .

↳ Proof of the Chinese Remainder Theorem

- Proof of *Existence of Solution*: $x \equiv a_i \pmod{m_i}$, $i=1, 2, \dots, k$. $m_1, m_2, \dots, m_k$ are pairwise coprime positive integers:
- ① Compute $M = m_1 m_2 \dots m_k$.
 - ② Define $M_i = M / m_i$, Since M_i is the product of all moduli except m_i . Since the moduli are pairwise coprime, we have $\gcd(m_i, M_i) = 1$.
 - ③ Because M_i and m_i coprime, by the theorem on the existence of modular inverses, there exists an integer m_i' such that $M_i m_i' \equiv 1 \pmod{m_i}$.
 - ④ Construct a partial solution for each congruence as $x_i = a_i M_i m_i'$, where each x_i satisfies the congruence $x \equiv a_i \pmod{m_i}$, but not affect m_j ($j \neq i$).
 - ⑤ The final solution x is the sum of all x_i , $x = \sum_{i=1}^k x_i$.

Proof of the Chinese Remainder Theorem

■ Proof of *Existence of Solution*: $x \equiv a_i \pmod{m_i}$, $i=1, 2, \dots, k$. $m_1, m_2, \dots, m_k$ are pairwise coprime positive integers:

⑥ To show that this x satisfies each individual congruence $x \equiv a_i \pmod{m_i}$:

Let $x = \sum_{j=1}^k x_j$, two cases on $x \pmod{m_i}$:

(a) For $j=i$, $x_i = a_i M_i m_i'$, and the result modulo m_i is a_i , since $M_i m_i' \equiv 1 \pmod{m_i}$.

(b) When $j \neq i$, because M_j includes m_i as a factor (since all m_i are pairwise coprime), we have $M_j m_j' \equiv 0 \pmod{m_i}$.

Thus, all x_j for $j \neq i$ contribute 0 modulo m_i , and only x_i determines the value of $x \pmod{m_i}$, yielding $x \equiv a_i \pmod{m_i}$.

Proof of the Chinese Remainder Theorem

- Proof of *Uniqueness of the Solution*: To prove that the solution x is unique modulo $M = m_1 m_2 \dots m_k$, it is sufficient to show that any two solutions x and y are congruent modulo M , i.e., $x \equiv y \pmod{M}$.
- ① Suppose there exist two solutions x and y that satisfy all the congruences in the system. That is, for every i , $x \equiv a_i \pmod{m_i}$ and $y \equiv a_i \pmod{m_i}$.
- ② Since both x and y yield the same remainder modulo each m_i , it follows that $x \equiv y \pmod{m_i}$. This implies that each m_i divides $(x - y)$.
- ③ Because $m_1, m_2, \dots, m_k$ are pairwise coprime, it follows from the properties of coprime integers that their product $M = m_1 m_2 \dots m_k$ also divides $(x - y)$.
- ④ $M \mid (x - y)$ means $x \equiv y \pmod{M}$, which proves that the solution is *unique modulo M* .

Chinese Remainder Theorem: Solving Linear Congruences

- Solving a System of Linear Congruences $x \equiv a_i \pmod{m_i}$, $i=1, 2, \dots, k$ where the positive integers $m_1, m_2, \dots, m_k$ are *pairwise coprime*.

- Solution: *Using the Chinese Remainder Theorem*.

- (1) Compute the product of all moduli: $M = m_1 m_2 \dots m_k$.
- (2) For each $i=1, 2, \dots, k$ compute $M_i = M / m_i$, $i=1, 2, \dots, k$.
- (3) Find the modular inverse of each M_i modulo m_i , denoted as M_i^{-1} .
- (4) Compute the final solution: $x = \sum_{i=1}^k a_i M_i M_i^{-1} \pmod{M}$.

By following the above steps, one can solve a system of linear congruences. In particular, when the moduli are pairwise coprime, the **Chinese Remainder Theorem** provides an efficient method for finding the solution.

Chinese Remainder Theorem: Solving Linear Congruences

- Example: Solve the "*Problem of the Unknown Quantity*," that is, find the positive integer solution to the following system of equations:
 $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$.

- Solution:

- ① $m_1=3$, $m_2=5$, $m_3=7$, $M=105$.
- ② $M_1=M/m_1=105/3=35$, $M_2=M/m_2=105/5=21$, $M_3=M/m_3=105/7=15$.
- ③ Solve $M_1^{-1} = 2$, $M_2^{-1} = 1$, $M_3^{-1} = 1$.
- ④ Final solutions $x = (2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1) \pmod{105} = 23$.
Solution is $105k + 23$, $k=0, 1, 2, \dots$. Least number is 23.

- Finding the Modular Inverse by Observation:

- ① Find the inverse of $M_1 = 35$ modulo $m_1 = 3$, i.e., find M_1^{-1} such that $M_1 \cdot M_1^{-1} \equiv 1 \pmod{3}$.
- ② ($35 \equiv 2 \pmod{3}$), we need to make $2 \cdot M_1^{-1} \equiv 1 \pmod{3}$.
- ③ By simple trial, we find that $M_1^{-1} = 2$ satisfies the condition.

- 8.4.1 Linear Congruences

Modular Inverses

- 8.4.2 The Chinese Remainder Theorem

- 8.4.3 Arithmetic Operations with Large Integers

↳ Modular Representation and Operations of Integers

- Let $m_1, m_2, \dots, m_k$ be integers greater than 1 and pairwise coprime, and define $m = m_1 m_2 \dots m_k$. For any integer $0 \leq x < m$, define $x_i = x \bmod m_i$, $i = 1, \dots, k$. $(x_1, x_2, \dots, x_k)$ is called the **modular representation of x** with respect to the moduli $m_1, \dots, m_k$, or simply the modular representation of x . It is denoted as: $x = (x_1, x_2, \dots, x_k)$.

- Modular Representation Operations

Let $x = (x_1, x_2, \dots, x_k)$, $y = (y_1, y_2, \dots, y_k)$, then

$$x + y = ((x_1 + y_1) \bmod m_1, (x_2 + y_2) \bmod m_2, \dots, (x_k + y_k) \bmod m_k).$$

$$x - y = ((x_1 - y_1) \bmod m_1, (x_2 - y_2) \bmod m_2, \dots, (x_k - y_k) \bmod m_k).$$

$$xy = (x_1 y_1 \bmod m_1, x_2 y_2 \bmod m_2, \dots, x_k y_k \bmod m_k).$$

Advantages of Modular Representation of Integers

Advantages of Representing an Integer x by Its Remainders Modulo a Set of Moduli:

- When an integer x is represented by its remainders with respect to a set of pairwise coprime moduli $m_1, m_2, \dots, m_k$, computations involving x can be carried out independently under each modulus.
- The results can then be combined using the **Chinese Remainder Theorem**.
- This approach improves **computational efficiency**, enhances **distributed processing capability**, and increases **algorithmic flexibility**.

■ **Example:** Let $m_1=9$, $m_2=7$, $m_3=5$, $m=9 \times 7 \times 5=315$, Arithmetic operations within the range of 0 to 314 can be performed using arithmetic modulo 9, 7, 5.

■ **Solve:** Let $x=20$, $y=13$, then $x=(2, 6, 0)$, $y=(4, 6, 3)$.

$$x+y=((2+4) \bmod 9, (6+6) \bmod 7, (0+3) \bmod 5)=(6, 5, 3).$$

$$x-y=((2-4) \bmod 9, (6-6) \bmod 7, (0-3) \bmod 5)=(7, 0, 2).$$

$$xy=(2 \times 4 \bmod 9, 6 \times 6 \bmod 7, 0 \times 3 \bmod 5)=(8, 1, 0).$$

Find the smallest positive integer solution:

$$\begin{cases} z \equiv 6 \pmod{9} \\ z \equiv 5 \pmod{7} \\ z \equiv 3 \pmod{5} \end{cases} \quad \begin{cases} z \equiv 7 \pmod{9} \\ z \equiv 0 \pmod{7} \\ z \equiv 2 \pmod{5} \end{cases} \quad \begin{cases} z \equiv 8 \pmod{9} \\ z \equiv 1 \pmod{7} \\ z \equiv 0 \pmod{5} \end{cases}$$

Compute as:

$$M_1=35, M_1 \equiv -1 \pmod{9}, M_1^{-1} = -1,$$

$$M_2=45, M_2 \equiv 3 \pmod{7}, M_2^{-1} = 5,$$

$$M_3=63, M_3 \equiv 3 \pmod{5}, M_3^{-1} = 2,$$

Then

$$x+y=(6 \times (-1) \times 35 + 5 \times 5 \times 45 + 3 \times 2 \times 63) \bmod 315 = 33.$$

$$x-y=(7 \times (-1) \times 35 + 0 \times 5 \times 45 + 2 \times 2 \times 63) \bmod 315 = 7.$$

$$xy=(8 \times (-1) \times 35 + 1 \times 5 \times 45 + 0 \times 2 \times 63) \bmod 315 = 260.$$

8.4 Linear Congruence Equations and the Chinese Remainder Theorem • Brief summary

Objective :

Key Concepts :



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- 8.1 Prime Numbers
- 8.2 Greatest Common Divisor and Least Common Multiple
- 8.3 Congruence
- 8.4 Linear Congruence Equations and the Chinese Remainder Theorem
- 8.5 Euler's Theorem and Fermat's Little Theorem

- Fermat's Little Theorem
- Euler's Totient Function
- Euler's Theorem

↳ Fermat's Little Theorem and Applications

■ Theorem 8.15 (Fermat's Little Theorem):

Let p be a prime number and a an integer such that a and p are coprime. Then: $a^{p-1} \equiv 1 \pmod{p}$.

■ Note:

- Furthermore, for any integer a , it holds that $a^p \equiv a \pmod{p}$.
- p is prime, a and p coprime, then $a^{p-1} - 1$ is divisible by p .
- $a^{p-1} \equiv 1 \pmod{p}$ is also $a * a^{p-2} \equiv 1 \pmod{p}$, $a^{p-2} \pmod{p}$ is the modular inverse of a .

↳ Fermat's Little Theorem and Applications

- Applications of Fermat's Little Theorem:
 - *Solving congruences* $ax \equiv b \pmod{p}$, When p is a prime, the theorem helps simplify the computation of powers modulo p .
 - *Primality testing*: While not absolutely reliable, Fermat's Little Theorem is effective in many cases for checking whether a number is prime.
 - *Proving a number is composite*:
If the theorem fails for a particular base a , it can be used to confirm that a number is not prime.
 - *To test whether a number p is prime*: If there exists any integer a ($1 < a < p$ and a is coprime to p) such that $a^{p-1} \pmod{p} \neq 1$, then p can be definitively identified as a **composite number**.
- Example: $2^{9-1} \equiv 4 \pmod{9}$, so we can conclude that 9 is composite.

↳ Euler's Totient Function and Its Computation

- **Euler's Totient Function $\phi(n)$** : It is the number of integers in the set $\{0, 1, \dots, n-1\}$ that are coprime to n .
- **Properties of $\phi(n)$** : If n is a prime number, then $\phi(n)=n-1$. If n is a composite number, then $\phi(n)<n-1$.
- **Formula for computing $\phi(n)$** : If the prime factorization of n is $n = p_1^{k_1} \cdot p_2^{k_2} \cdots p_r^{k_r}$, where $p_1, p_2, \dots, p_r$ are distinct prime numbers, then
$$\phi(n) = n \cdot \left(1 - \frac{1}{p_1}\right) \cdot \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_r}\right)$$
- **Example 1: Direct Counting Method** to Find $\phi(4)$.
 $\{0, 1, \dots, 3\}$, only 1 and 3 are coprime to 4, $\phi(4)=2$.
- **Example 2: Prime Factorization Method** to Find $\phi(12)$.
 The prime factorization of 12 is $2^2 \times 3$, $p_1=2$ and $p_2=3$,
 Therefore: $\phi(12) = 12 \cdot \left(1 - \frac{1}{2}\right) \cdot \left(1 - \frac{1}{3}\right) = 4$.

↳ Euler's Theorem

■ Theorem 8.16 (Euler's Theorem):

If a is coprime to n , then $a^{\phi(n)} \equiv 1 \pmod{n}$.

■ Note:

- Since $\phi(n)$ when n is a prime, Euler's Theorem becomes **Fermat's Little Theorem** in that case.
- Thus, **Euler's Theorem is a generalization of Fermat's Little Theorem**, and it can be applied to any positive integer n .
- When n is a composite number, **Euler's Theorem provides a way to find the modular inverse of a modulo n** , because $a^{\phi(n)-1} \equiv a^{-1} \pmod{n}$.

Objective :

Key Concepts :